



NYU

Center for Urban
Science + Progress

Flooding Simulation: New York City Ida event

Junyi Li

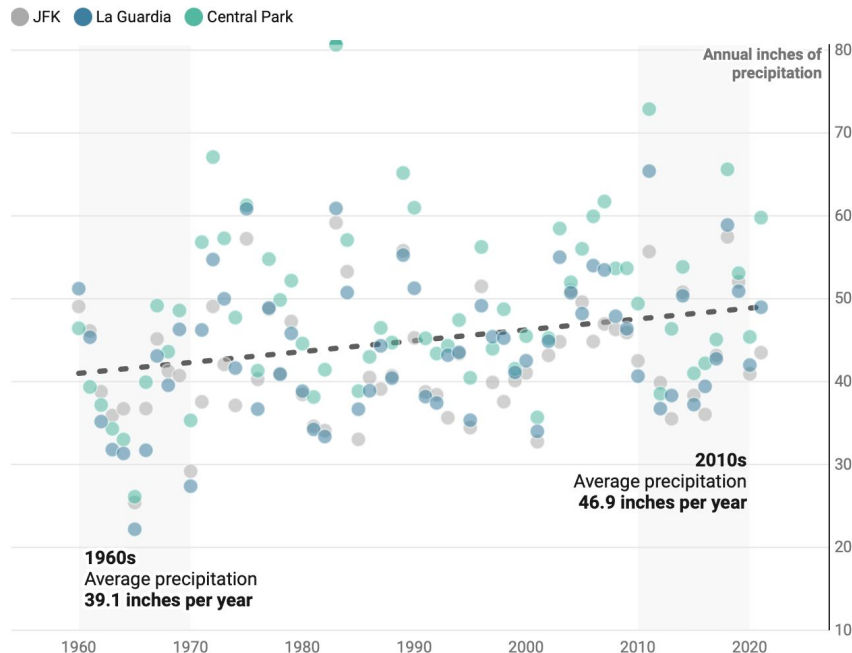
New York City's Pluvial flooding risk

Modeling is one of the most important tools for understanding where rainfall will generate flooding and what modifications are needed to prevent it.

– NYC GOV 2024 storm water analysis

<https://www.nyc.gov/assets/dep/downloads/pdf/water/stormwater/2024-stormwater-analysis-report.pdf>

NYC's annual precipitation has increased about 20% since 1960

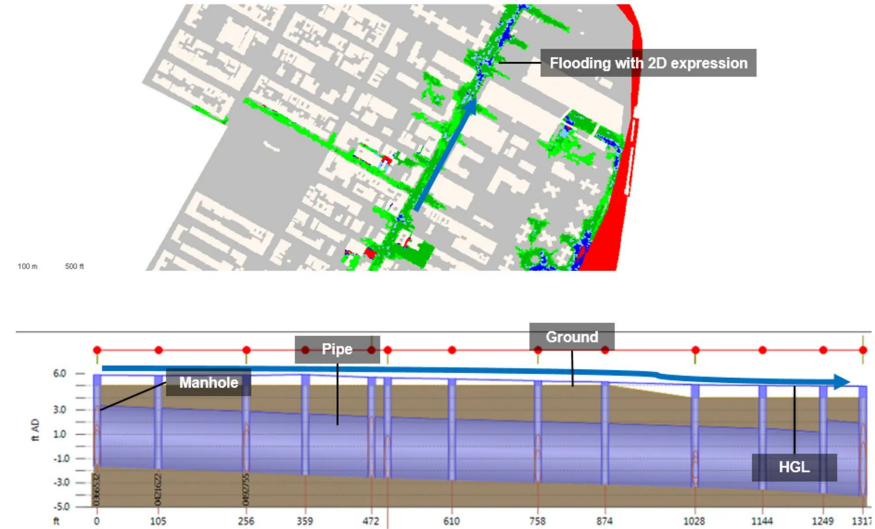


Source: NPCC • Get the data • Created with Datawrapper

Image: [NYC Open Data Story](#)

Hydraulic - hydrologic models

- Hydraulic - sewer network
- Hydrologic - surface water behavior
- Advantages
 - Accurate risk assessment and prediction
- Disadvantages
 - Requires drainage data
 - Long computational time
 - High cost



Paniconi, C., & Putti, M. (2015). Physically based modeling in catchment hydrology at 50: Survey and outlook. *Water Resources Research*, 51(9), 7090-7129.

Image: [Advanced 2D Hydraulic Modeling and LIDAR Integration for Resiliency Planning](#)

Cellular automata approach

- Flow Drivers: Slope between cells
- Water flows only from higher-elevation to lower-elevation adjacent NH cells per time step
- Advantages
 - Fast Flood Model
 - Suitable for parallelization
- Disadvantage
 - Didn't consider the sewer network flow
 - Didn't include hydrological losses, e.g. infiltration
 - Computation time increases with study area: 3 min for 405,700 m²; estimated significantly longer for NYC (783.8 million m²).

Ghimire, Bidur & Chen, Albert & Guidolin, Michele & Keedwell, Edward & Djordjević, Slobodan & Savic, Dragan. (2003). Formulation of a fast 2D urban pluvial flood model using a cellular automata approach. Journal of Hydroinformatics. 15. 676-686. 10.2166/hydro.2012.245.

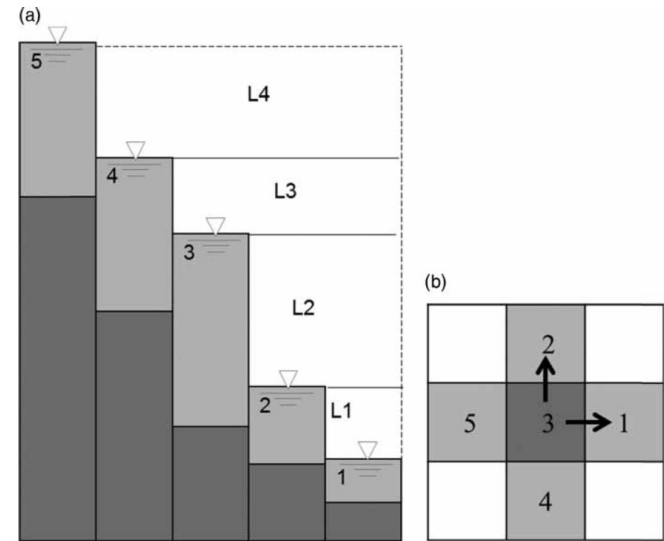


Figure 1 | (a) Cells ordered in NH according to their ranks, L1–L4 are layers of free spaces between the water levels of two cells that are available within NH for water distribution, the numbers shown are cell ranks. In this diagram the ground level for each cell is shown in dark grey and the water level light grey, (b) an example of outflow fluxes (shown by arrows) from the central cell having rank 3 to its neighbouring cells.

Image: [Formulation of a fast 2D urban pluvial flood model using a cellular automata approach](#)

Weighted Cellular automata approach

- The parallel model implementation runs on multi-core CPUs and graphics card GPUs
- Distribute this volume to downstream cells proportionally to their weights.
- Advantages
 - Speed Up use GPUs
 - Less sensitive to the size of the time step
- Disadvantage
 - **Sensitive to the grid resolution**
 - **Didn't include hydrological losses, e.g. infiltration**
 - **Decrease in the computational speedup at higher resolutions**

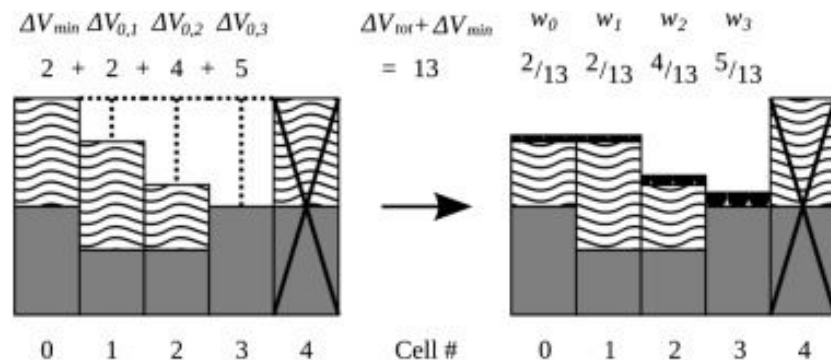
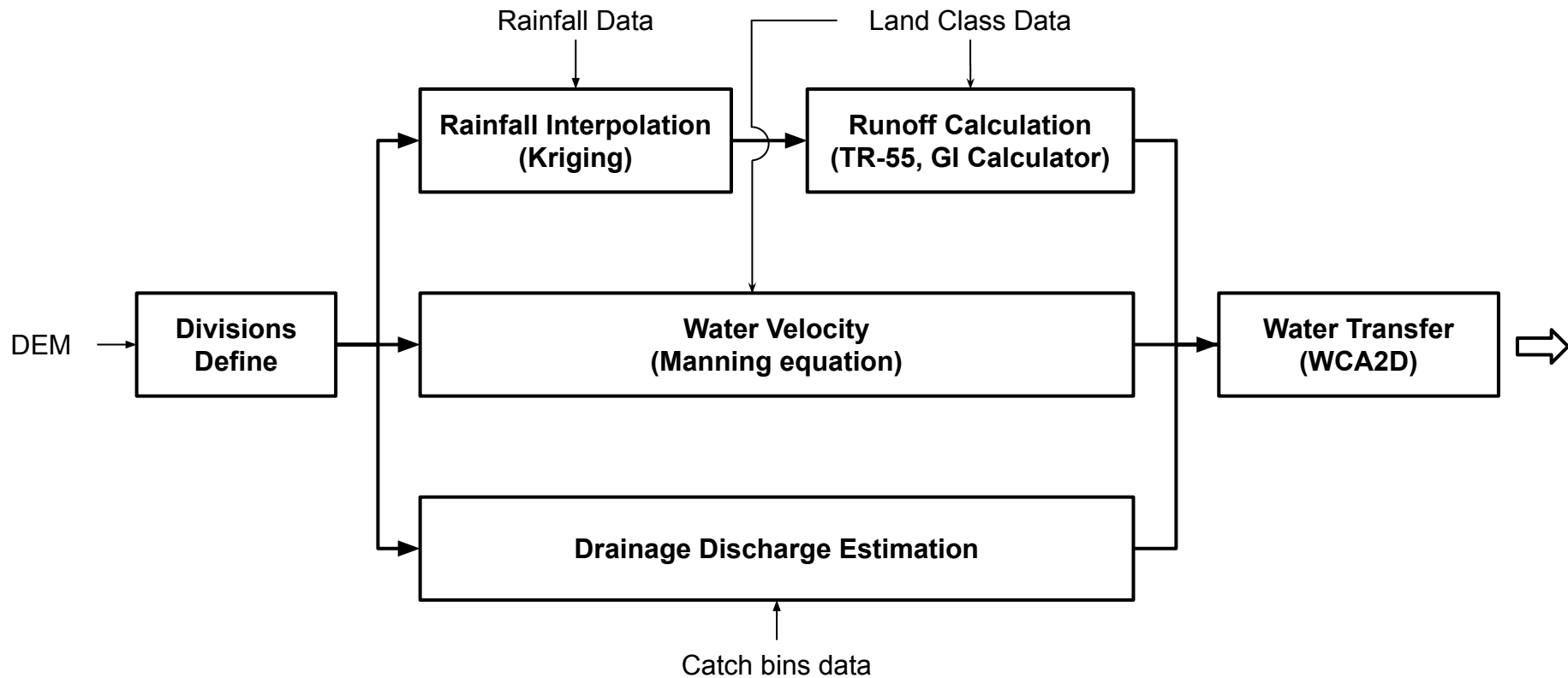


Image: [Example of intercellular-volume computation.](#)

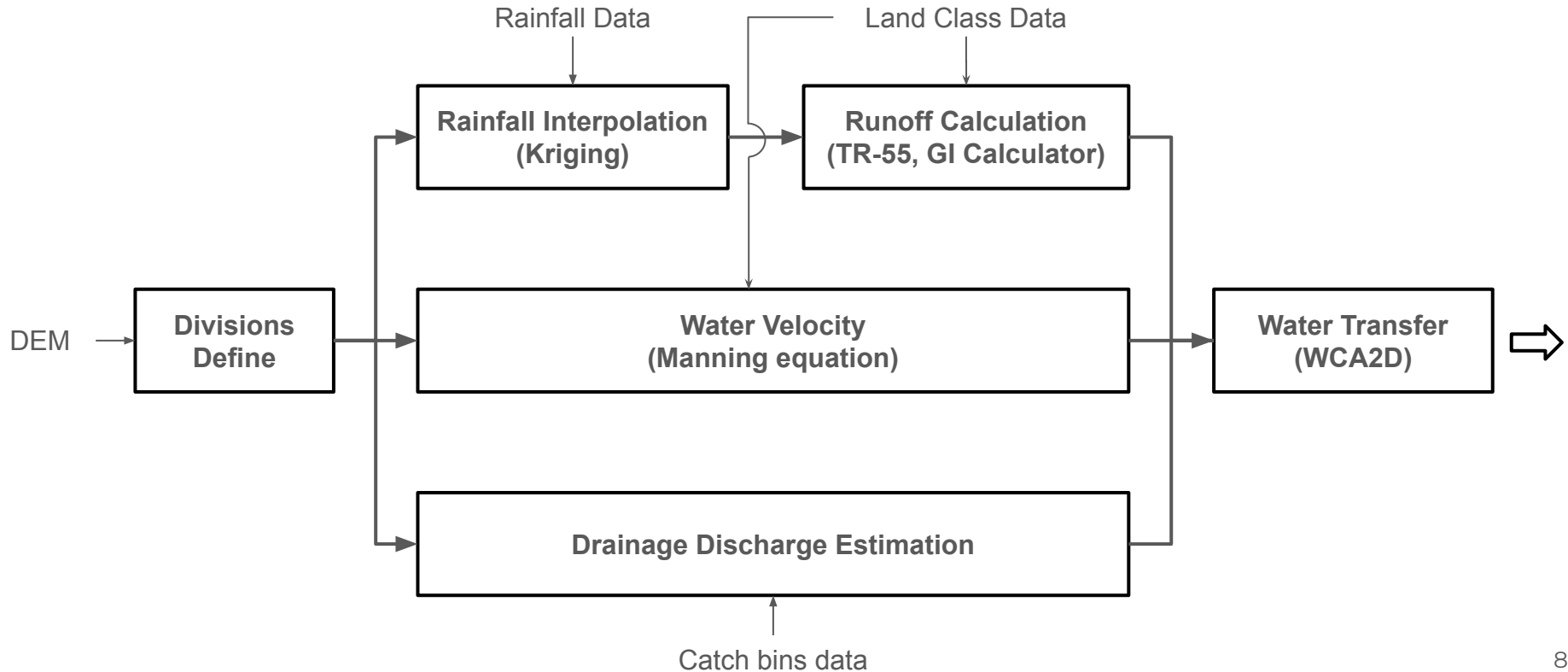
Research Goal

Develop a computationally efficient rainfall-induced flood simulation model to accurately predict rainfall distribution and flood depth at target locations, with the following key features:

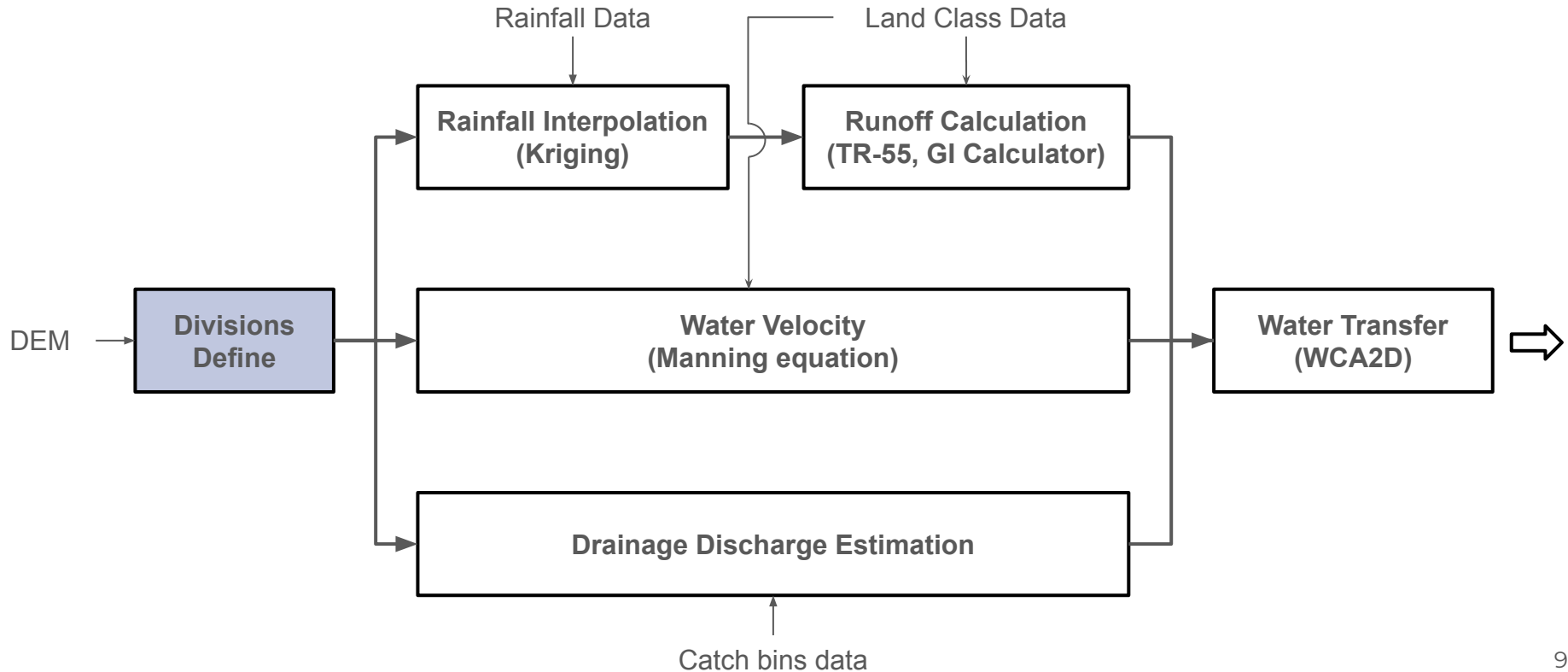
- Handles **non-uniform rainfall inputs** across the study area
- Incorporates **hydrological losses** (e.g., infiltration, retention)
- Estimates **sewer system drainage capacity** with limited data availability
- Integrates a **Green Infrastructure (GI) calculator** to assess mitigation impacts



The proposed model



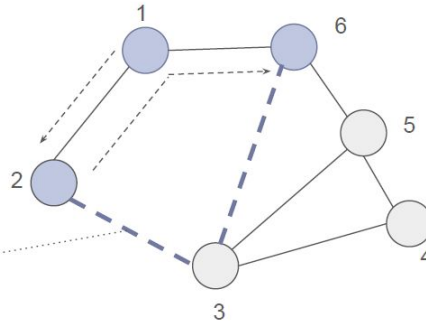
The proposed model



Divisions Define



{ (Ploy_ID_1, Ploy_ID_2) : N1,
 (Ploy_ID_3, Ploy_ID_6) : N2,
 (Ploy_ID_3, Ploy_ID_2) : N3,
 (Ploy_ID_4, Ploy_ID_5) : N4, ... }



[1,2,3] $\leftarrow$ current stack
 Areas = $A1 + A2 + A3$
 $> \text{threshold}$

Start a new stack
 [1,2]
 [3] $\leftarrow$ current stack

Remove link if:

- avg_elev of shared edge larger than both of basins ($N3 > M3$ and $N3 > M2$), or
- both basin areas larger than threshold A ($A1 > A$ and $A3 > A$)

Overall Rule: If the shared boundary is lower, combine the two basins

- Basin delineation by ArcGIS Pro basin tool.
- Intercept and Zoning Statistic.
- Depth First Search Algorithm.
- Merge the Division.

Divisions Define

Overall Rule: If the shared boundary is lower, combine the two basins

Dictionary of Nodes (basin) : Area and Average elevation of boundary

{ Ploy_ID_1 : A1, Ploy_ID_2 : A2, ..., Ploy_ID_6 : A6 }

{ Ploy_ID_1 : M1, Ploy_ID_2 : M2, ..., Ploy_ID_6 : M6 }

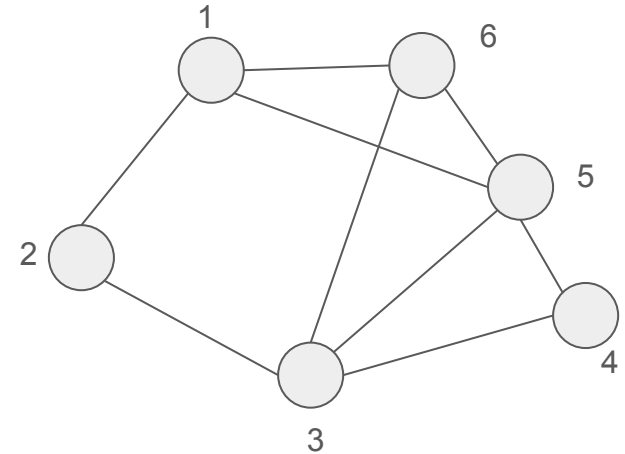
Dictionary of Edge (Shared boundary) : Average elevation

{ (Ploy_ID_1, Ploy_ID_2) : N1, (Ploy_ID_3, Ploy_ID_6) : N2,

(Ploy_ID_3, Ploy_ID_2) : N3, (Ploy_ID_4, Ploy_ID_5) : N4, ... }

Represent polygons and their connections as a graph

- **Nodes** = Basins.
- **Edges** = Shared boundaries between basins



Divisions Define

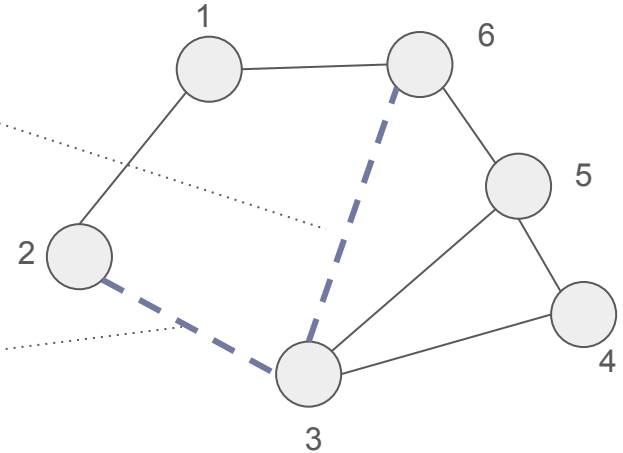
Remove link if both basin areas
larger than 20,000,000 sqft

$A1 > 2000000$ and $A3 > 2000000$

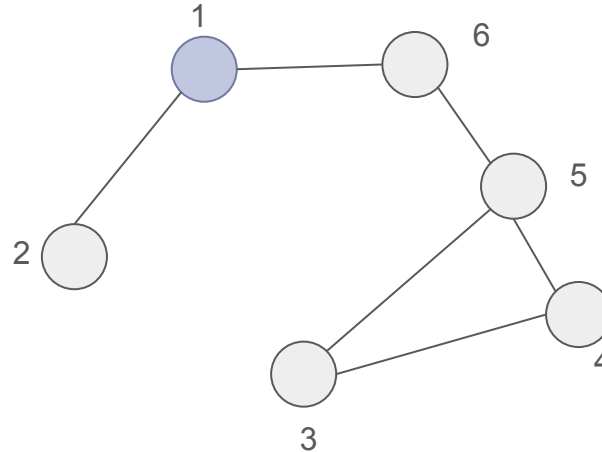
$\{ (\text{Ploy_ID_1}, \text{Ploy_ID_2}) : N1, (\text{Ploy_ID_3}, \text{Ploy_ID_6}) : N2, \\ (\text{Ploy_ID_3}, \text{Ploy_ID_2}) : N3, (\text{Ploy_ID_4}, \text{Ploy_ID_5}) : N4, \dots \}$

Remove link if avg_elev of shared edge
larger than both of basins

$N3 > M3$ and $N3 > M2$



Divisions Define

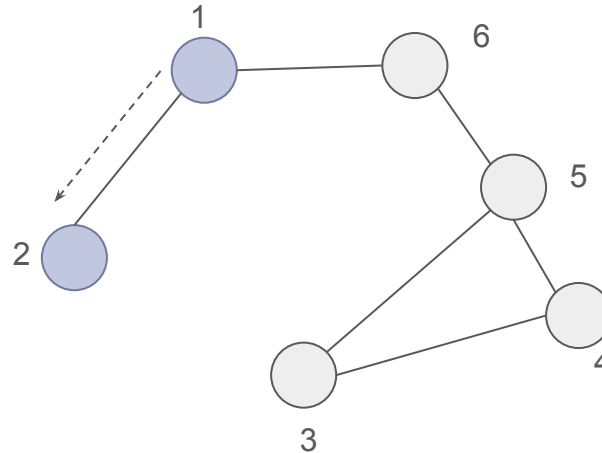


[1] $\leftarrow$ current stack
Areas = A1

Initialize with node 1
because basin 1 is the smallest
among all unvisited nodes

- Each nodes could have two stages: visited or unvisited
- Let the left nodes always have smaller areas than the right ones
- Initialized a new stack with the unvisit node with smallest areas
- Travel through unvisited node with DFS, until the total areas of current stack exceeded area limit of 10,000,000 sqft

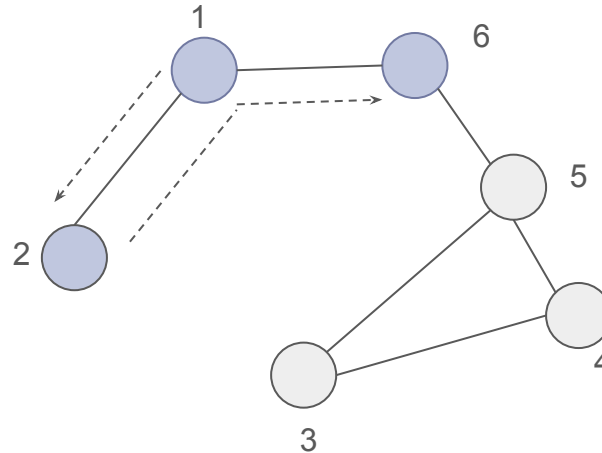
Divisions Define



[1,2] ← current stack
Areas = $A_1 + A_2$
< threshold

- Each nodes could have two stages: visited or unvisited
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Divisions Define



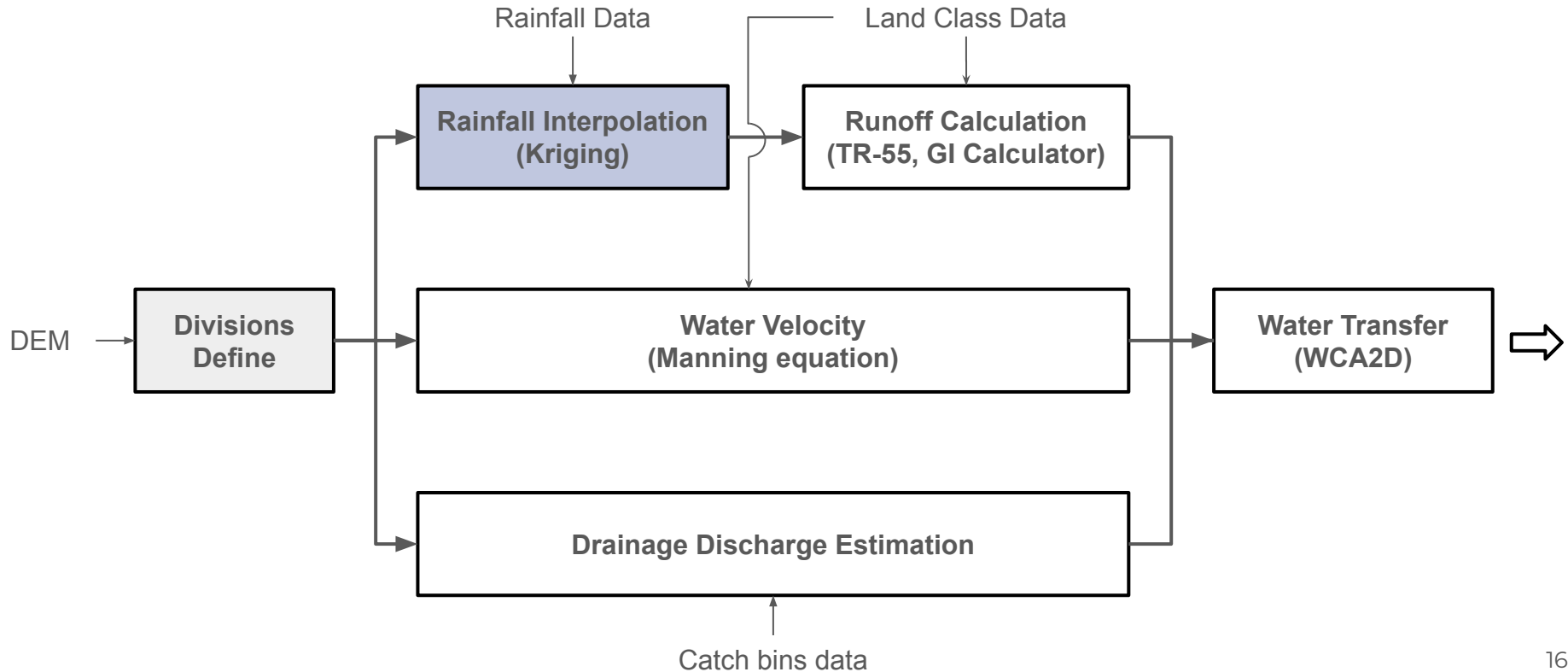
[1,2,3] ← current stack
 Areas = $A_1 + A_2 + A_3$
> threshold

Start a new stack

[1,2]
 [3] ← current stack

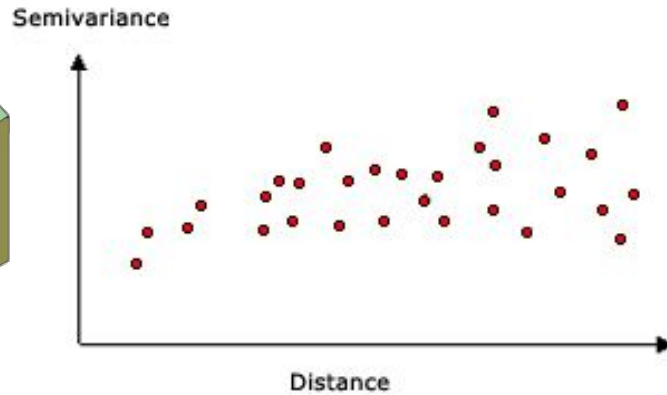
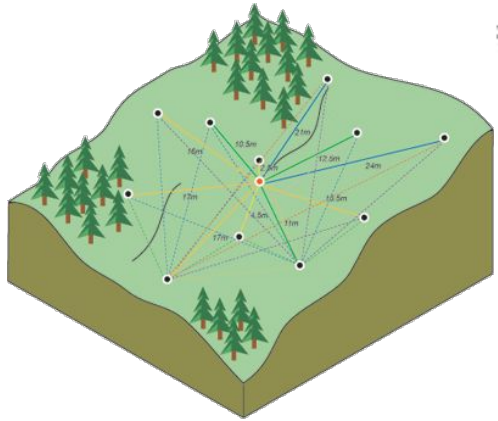
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The proposed model



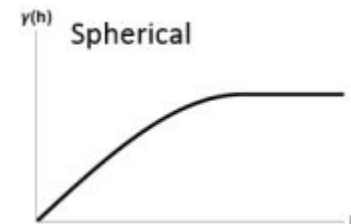
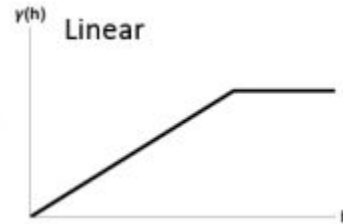
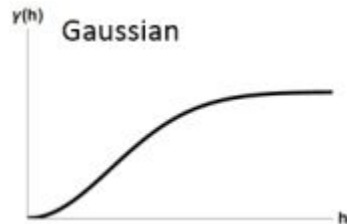
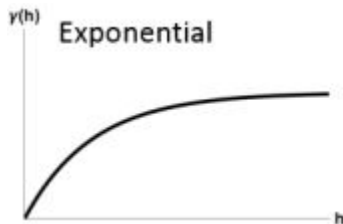
Rainfall Interpolation

$$\text{Semivariogram}(\text{distance}) = 0.5 * \text{average}((\text{value}_i - \text{value}_j)^2)$$

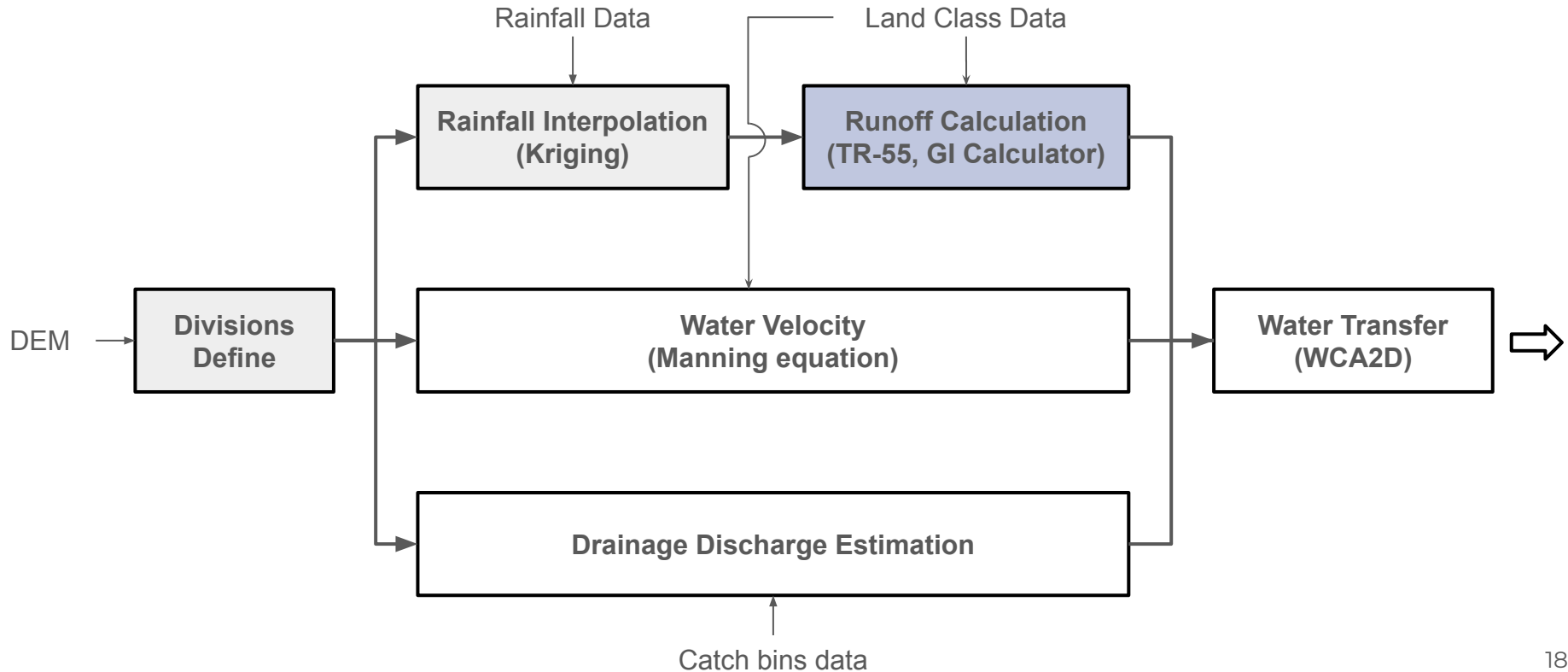


- Nearby points are given more weight than those farther away.
- Two-Step Process: Variogram Modeling and Interpolation

Image: ArcGIS Pro Documentation

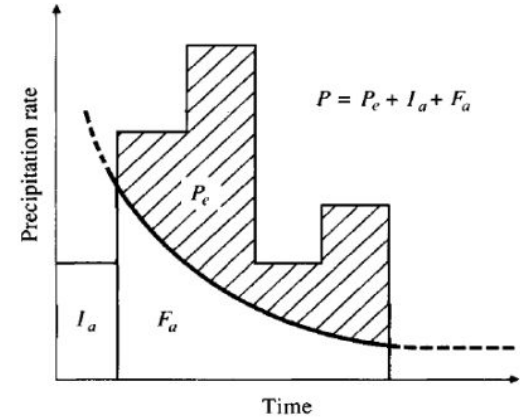


The proposed model



Runoff Calculation - SCS Runoff Method

$$Q = \begin{cases} 0 & \text{for } P \leq I_a \\ \frac{(P - I_a)^2}{P - I_a + S} & \text{for } P > I_a \end{cases} \quad \begin{aligned} S &= \frac{1000}{CN} - 10 \\ I_a &= 0.2S \end{aligned}$$



- Q is runoff ([L]; in)
- P is rainfall ([L]; in)
- S is the potential maximum soil moisture retention after runoff begins ([L]; in)
- I_a is the initial abstraction ([L]; in), or the amount of water before runoff, such as infiltration, or rainfall interception by vegetation
- CN has a range from 30 to 100; USDA indicate the runoff curve numbers for characteristic land cover descriptions and a hydrologic soil group

Green Infrastructure Calculator

Green infrastructure can mitigate flood risk by slowing and reducing stormwater runoff and protecting floodplains.

- Increase the initial abstraction by storage the water

$$I_a = \lambda * S_{sith\ gi}$$

- Decrease potential maximum retention after runoff begins

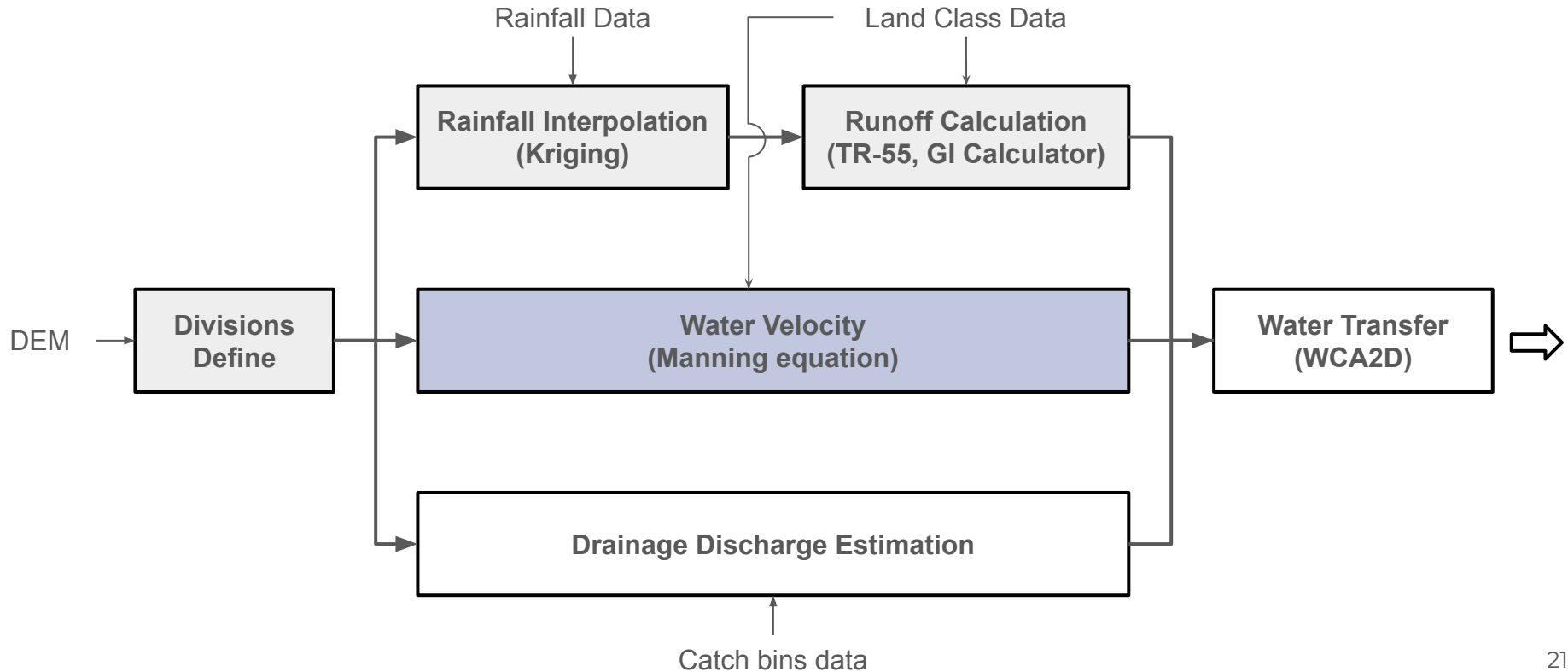
$$S_{with\ gi} = \frac{V_{with\ gi}}{A_{lc}} = \frac{S_{lc} * (A_{lc} - A_{gi}) + V_{gi}}{A_{lc}}$$



Image: [Green Values Calculator Methodology](#)

BMP	Category	Unit	Installed on
Green Roof	Roof Water Capture	ft ²	roof
Rain Barrel	Roof Water Capture	number	none
Cistern	Roof Water Capture	number	none
Foundation/Perimeter Drain	Roof Water Capture	ft	any
Rain Garden	Roof Water Redirection	ft ²	any
Planter Boxes	Roof Water Redirection	planters	any
Drywell	Roof Water Redirection	number	none
Trees	Landscaping	trees	any
Amended Soil	Landscaping	ft ²	any
Bio-Swales	Landscaping	ft	any
Urban Farming/Gardening	Landscaping	ft ²	any
Vegetation Filter Strip	Landscaping	ft	any
Native Vegetation	Landscaping	ft ²	any
Parking Lot Swales	Directing Runoff	ft	parking
Roadside Swales	Directing Runoff	ft	any
Permeable Patio	Permeable Paving	ft ²	patio
Permeable Parking	Permeable Paving	ft ²	parking
Permeable Sidewalks	Permeable Paving	ft ²	sidewalk
Permeable Driveway	Permeable Paving	ft ²	drive
Permeable Streets	Permeable Paving	ft ²	street

The proposed model



Water Velocity - Manning Equation

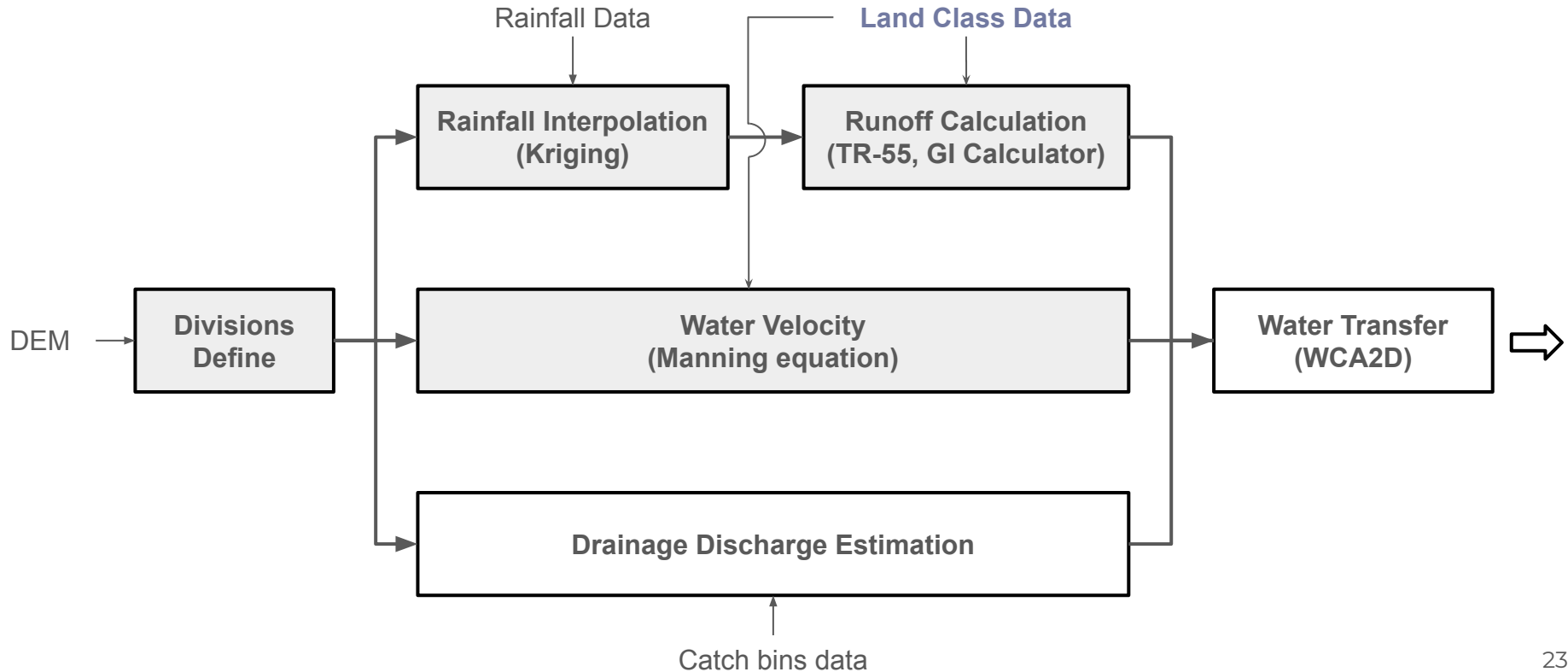
$$v = \frac{1.43}{n} R^{2/3} S^{1/2}$$

- v is average flow velocity (ft/s) for one division
- n is Manning's roughness coefficient for one division weighted by area
- R is hydraulic radius (ft) → for sheet flow, often approximated as flow depth
- S is average slope (ft/ft) for one division, Slope at a cell computed by weighted differences between neighboring elevations:

$$S = \sqrt{(dZ/dx)^2 + (dZ/dy)^2}$$

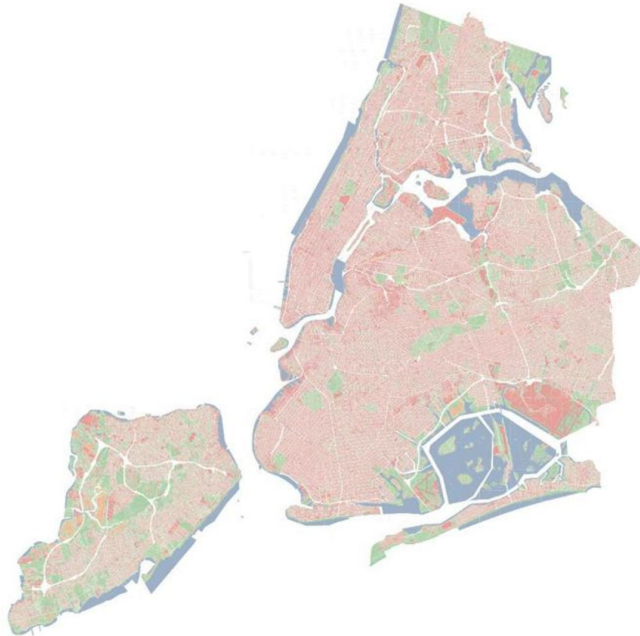
- Where dZ/dx and dZ/dy are gradients in x and y direction using a 3x3 window.(sobel filter)

The proposed model



Land Cover Data

DEP's Citywide Parcel-Based Impervious Area
LCpp (land cover classification per parcel)



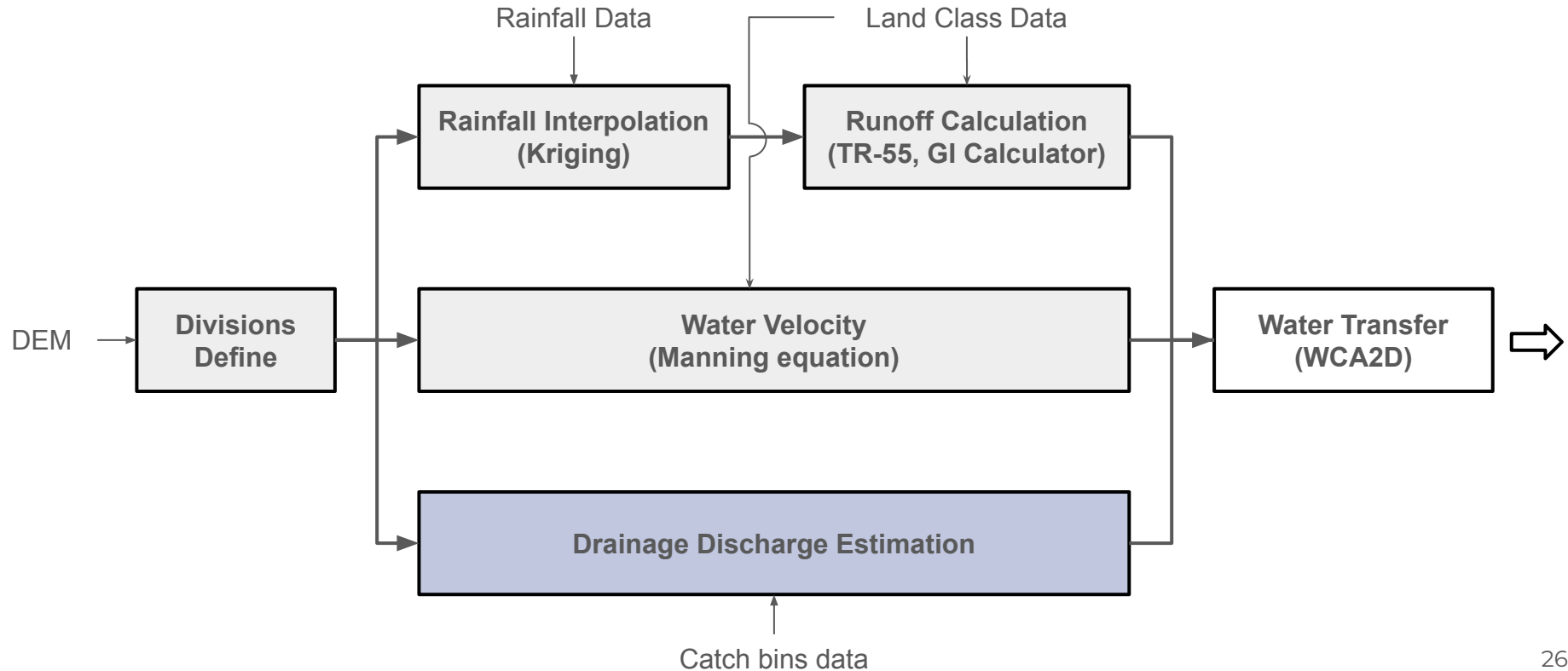
Land Cover Class	C-Value Range	Level of Imperviousness
1. Metal	> 0.98	Impervious
2. Rubber		
3. Wood		
4. Concrete	0.85-0.98	
5. Roof	0.85-0.95	
6. Asphalt	0.8-0.98	
7. Brick Paver		
8. Rock		
9. Solar Panel	N/A	
10. Pool		
11. Water		
12. Gravel	0.25-0.85	Semi-Pervious
13. Synthetic Turf	0.25-0.7	
14. Bare Soil	0.15-0.5	
15. Sand	0.3-0.5	Pervious
16. Grass	0-0.35	
17. Bush		
18. Tree		
19. Open Water	N/A	N/A

Image: NYC DEP

Land Cover Data

Land Cover Class	Manning's n (range)	Reference	TR-55 categories
asphalt			
concrete			
solar panel	0.011 – 0.013		
metal		Chow (1959); NRCS	impervious area
roof			
brick paver	0.012 – 0.015		
rock	0.025 – 0.035		
synthetic turf	0.030 – 0.035	ASCE (manual); engineering estimates	
wood	0.015 – 0.018	Chow (1959)	raised deck
grass	0.10 – 0.15		lawn in good condition
bush	0.20 – 0.40		
tree	0.30 – 0.50		shrub and bushes
bare soil	0.02 – 0.05	Chow (1959); NRCS	
sand	0.02 – 0.04		open
gravel	0.02 – 0.035		
pool			
open water	0.025 – 0.035	Chow (1959); USACE	water
water			

The proposed model



Estimate Discharge through drainage system

- New York City's sewer system is only designed to manage 1.75 inches of rain per hour.
- For each division: weighted by the number of catch bins in each division

Assumption: Each basin is designed to manage the same rain volume

$$D_i = \frac{(1.75 \cdot A)}{N} \cdot \frac{n_i}{A_i}$$

A: Total areas of New York City

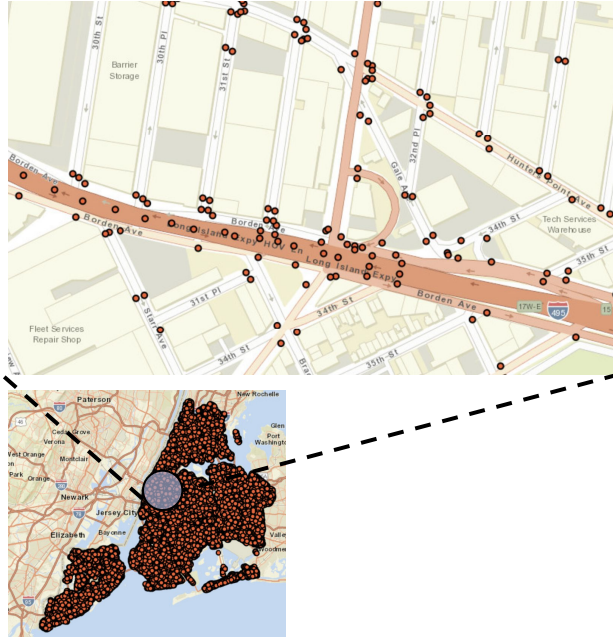
A_i: Areas of division i

N: Total number of catch bins in NYC

n_i: Number of catch bins in division i

D_i: the maximum Drainage system discharge

Catch Basins Data



UniqueID	Number_of_Basins	Area
1	1197	10677822
2	4465	20728215
4	170	890164.8
5	756	3778321
6	3019	17916911
8	480	2120583
9	57	173543.5
10	377	2274328
11	72	841236.5
12	379	2090458
14	166	497109.2
15	2352	10557484
16	287	1151064
17	136	496360.7
18	11	333392.7
19	682	2368672
20	139	924519.7
21	1	139626.9
22	50	632423.8
23	244	1383693
24	818	3786865
25	117	663399.9

Catch basins form part of DEP's Collection System Network and are available as an open dataset.

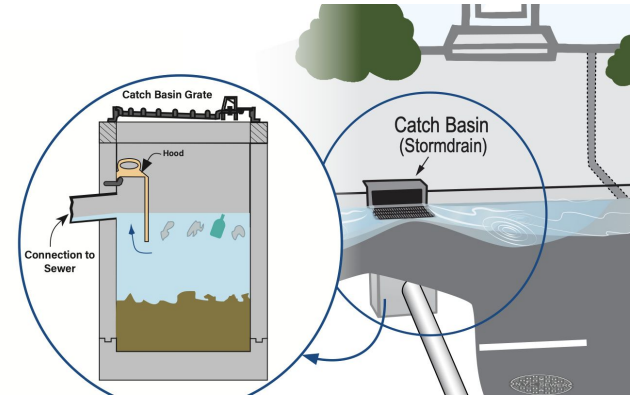
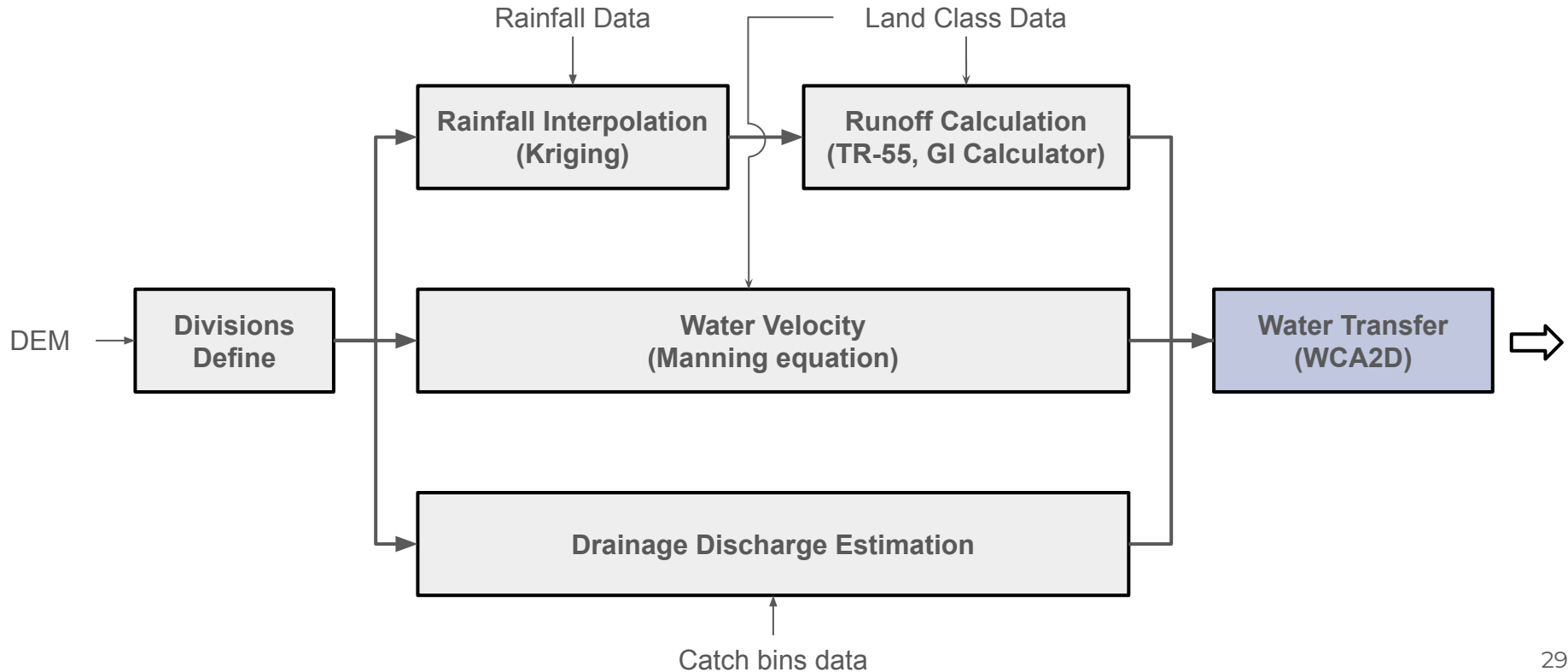


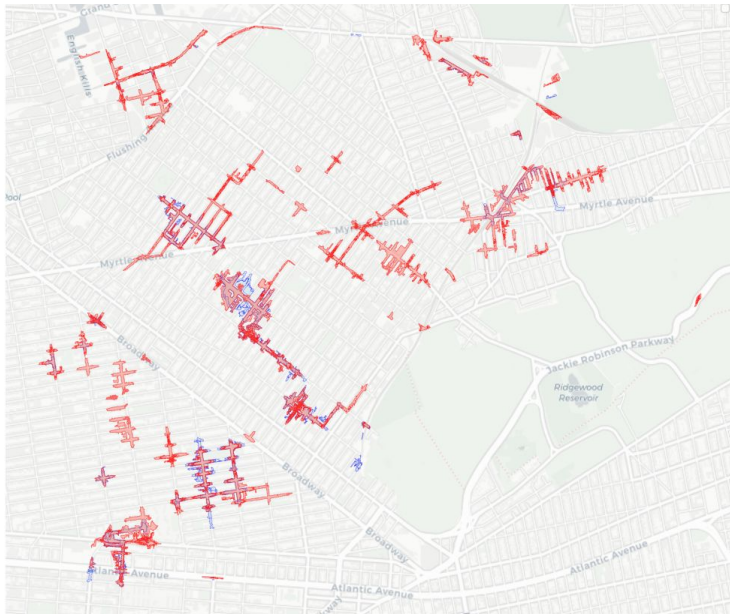
Image: NYC DEP

The proposed model



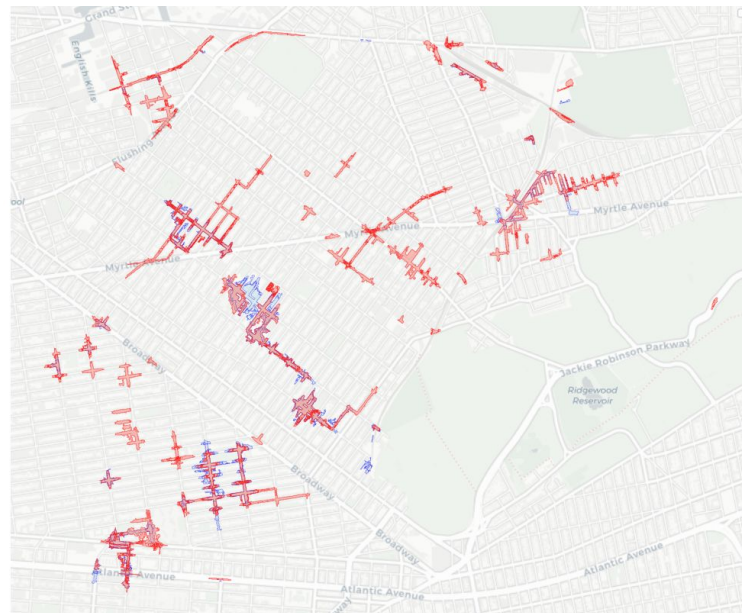
Intercellular-volume computation

CPU: CA2D model



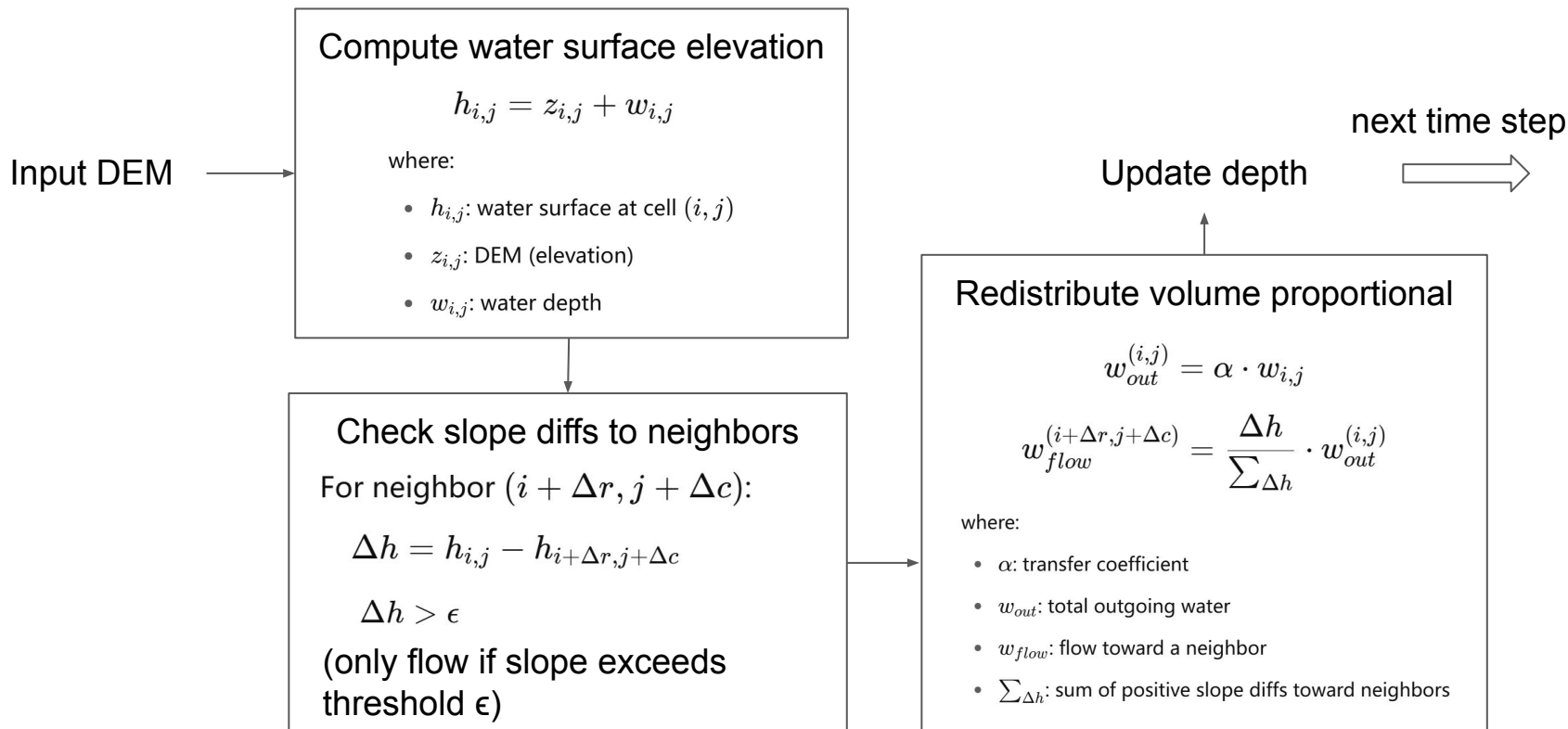
True Positive Rate (TPR): 0.7894
False Discovery Rate (FDR): 0.6388
Time: 3 min 7s

GPU: WCA2D model

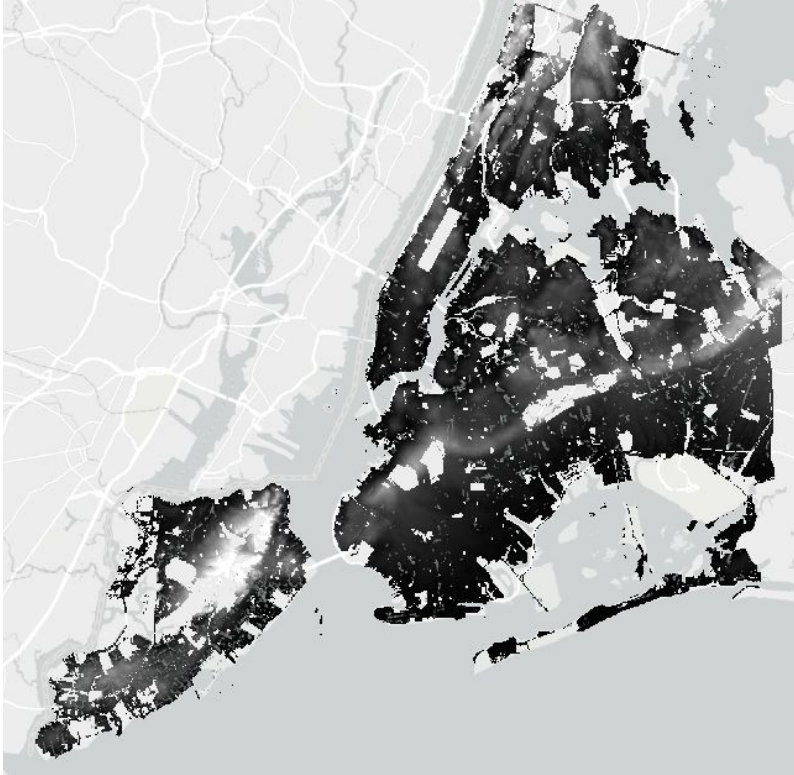


True Positive Rate (TPR): 0.7307
False Discovery Rate (FDR): 0.6097
Time: 1 min 32s

Inter-cellular-volume computation



Study Area: New York City



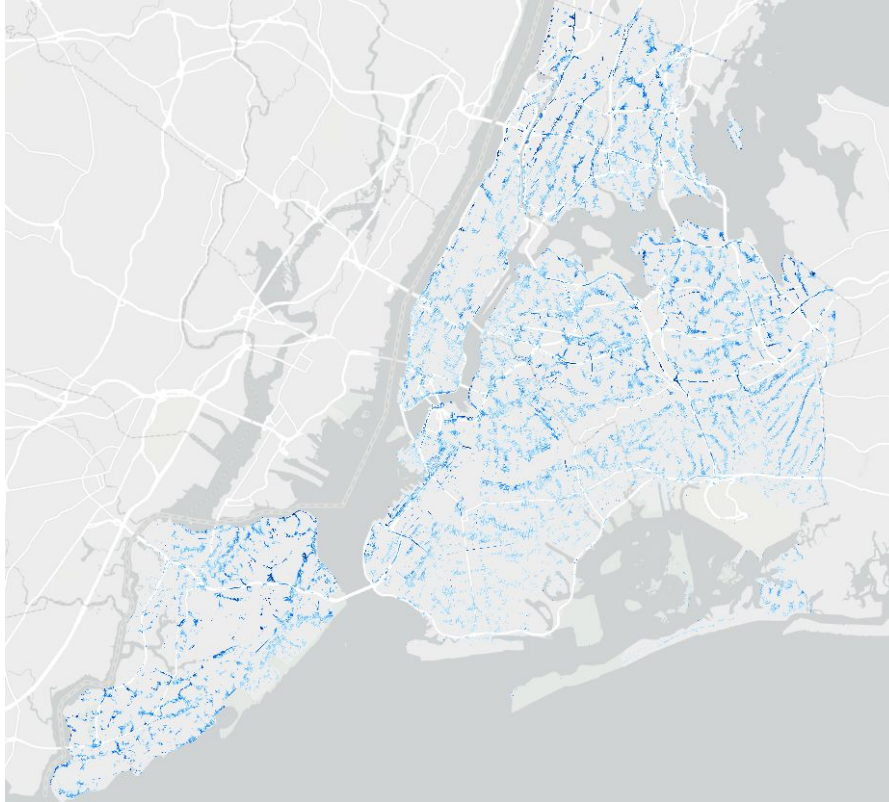
- Area: **783.8 million m²**
- Resolution: 14925 * 15128 * 10 foot (~ 3 meters)
- Some areas of the City were masked out because there was insufficient information about their drainage infrastructure to represent the flood risk.
 - Open Space Parks from the NYC Planimetric Database $\geq 100,000$ sqft,
 - Non-Residential and Non-Commercial PLUTO Lots $\geq 250,000$ sq ft,
 - PLUTO lots that intersected railway infrastructure.

Divisions



- 106,421 to 135 Divisions
- **Average area 53.43 million sq ft**

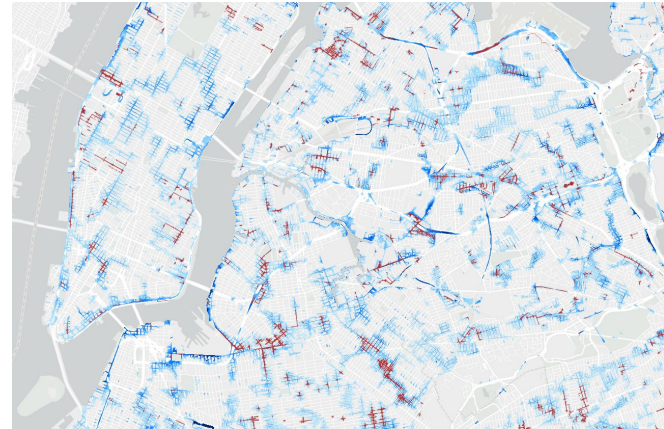
Scenario 1: 2.13 inch rainfall with current SLR



The TPR and FDR metrics, also respectively called probability of detection (hit rate) and false alarm ratio (Bennett et al., 2013), are used to compare the inundated area predicted by the two models.

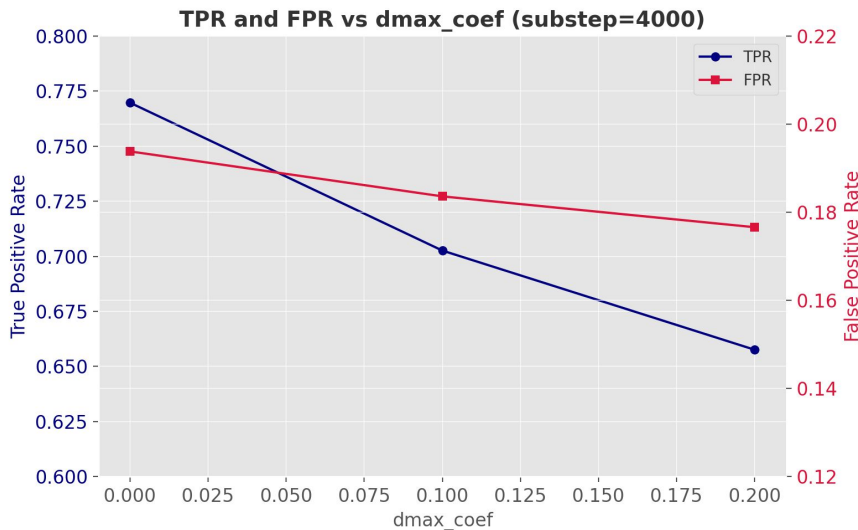
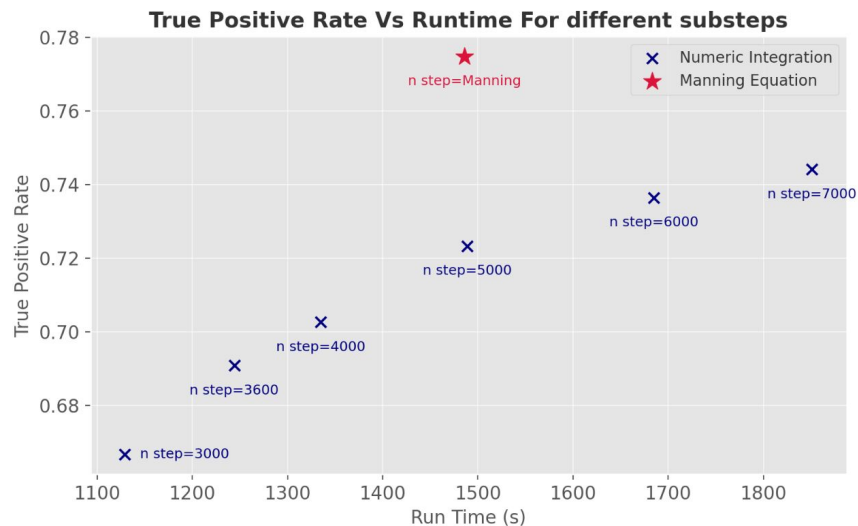
$$TPR = TP / (TP + FN) \quad 79.66\%$$

$$FDR = FP / (TP + FP) \quad 20.09\%$$

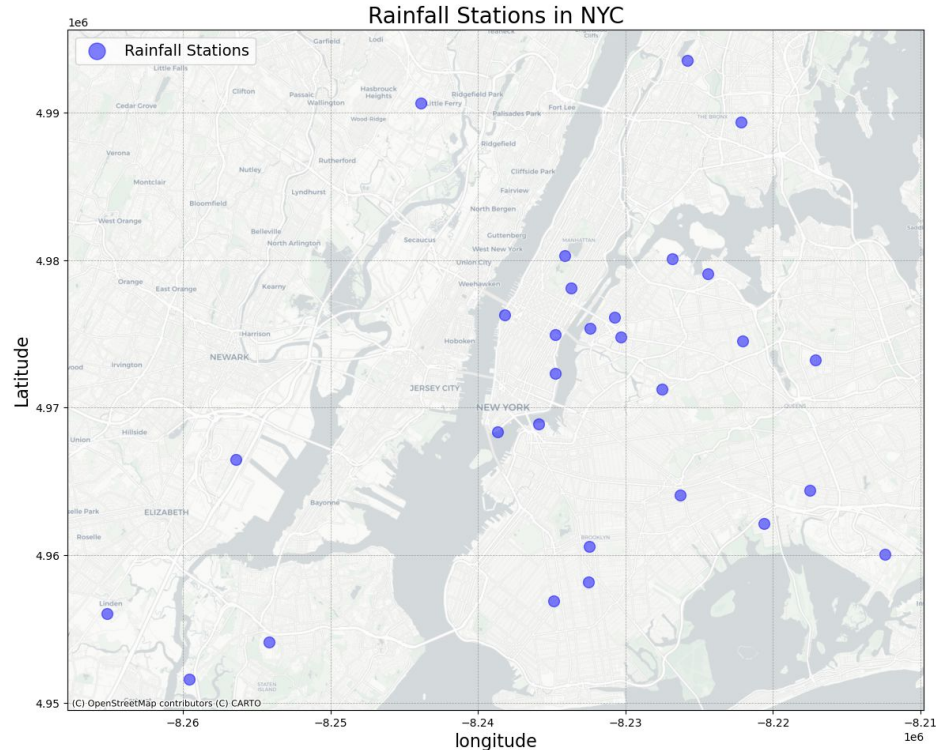


Scenario 1: Parameter tuning

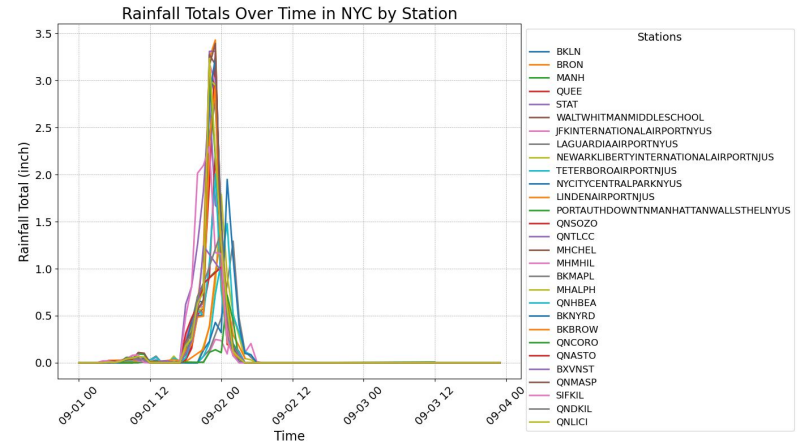
- Uses the Manning equation to compute a spatially-varying time step per cell increase the model accuracy and reduce runtime.
- Introduces an empirical maximum discharge (drange) for each drainage element, preventing overprediction. Lowers FDR while maintaining or slightly boosting TPR.



Scenario 2: Ida Rainfall

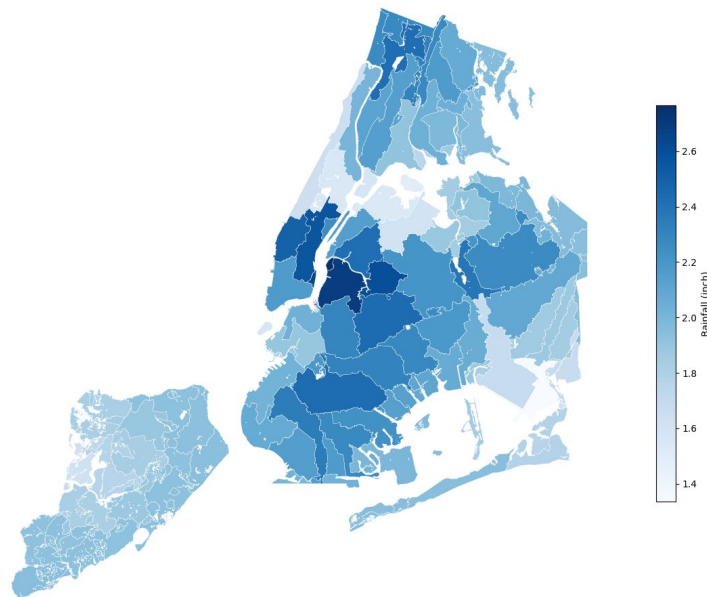


- The remnants of Hurricane Ida produced significant rainfall over the five boroughs of NYC on September 1st, 2021.
- Record-breaking 3.5 inches per hour of rainfall fell in some areas. Fourteen people died from causes related to Ida.



Scenario 2: Ida

Rainfall Distribution at 2021-09-01 23:00:00



Data	# of stations	R ²	RMSE (48 hour)	MAE (48 hour)
Ida event	29	28.62%	2.26 inch	1.93 inch

- Testset: National Weather service 24/48 h rainfall amount
- RMSE: 2.26 inch for 48h
- Std of test data: 1.53 inch for 48h

Scenario 2: Ida



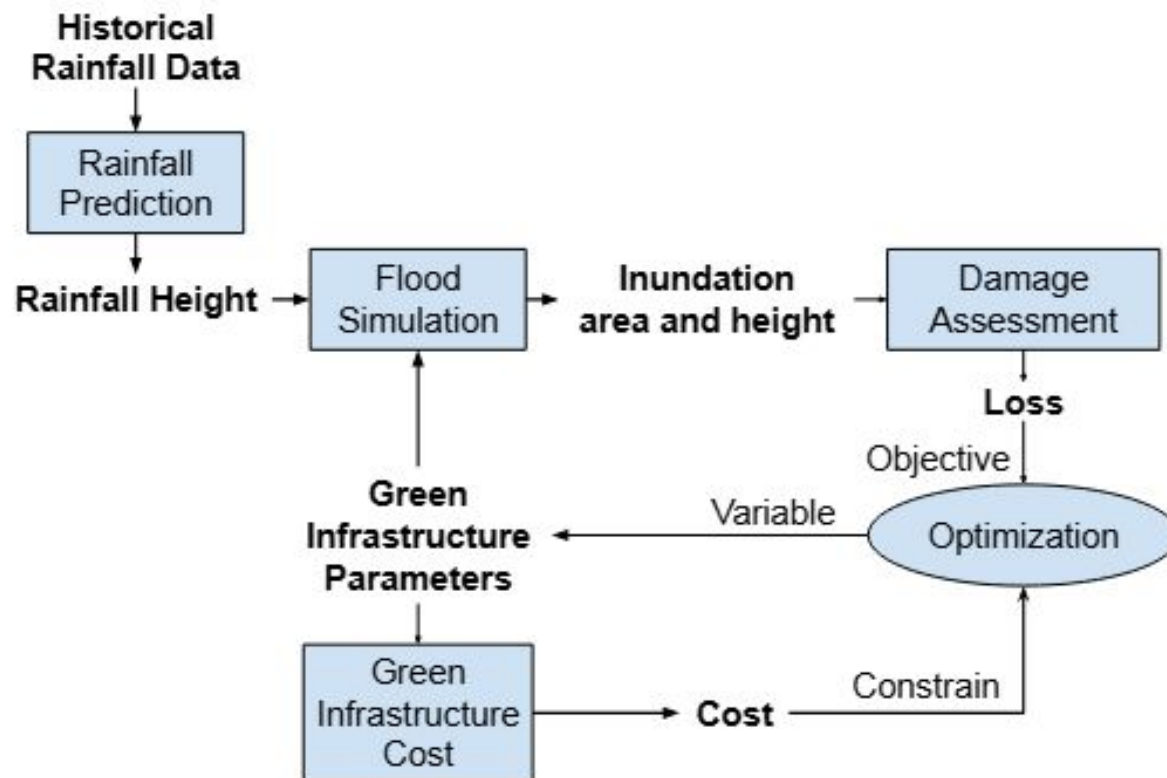
Scenario 2: Ida

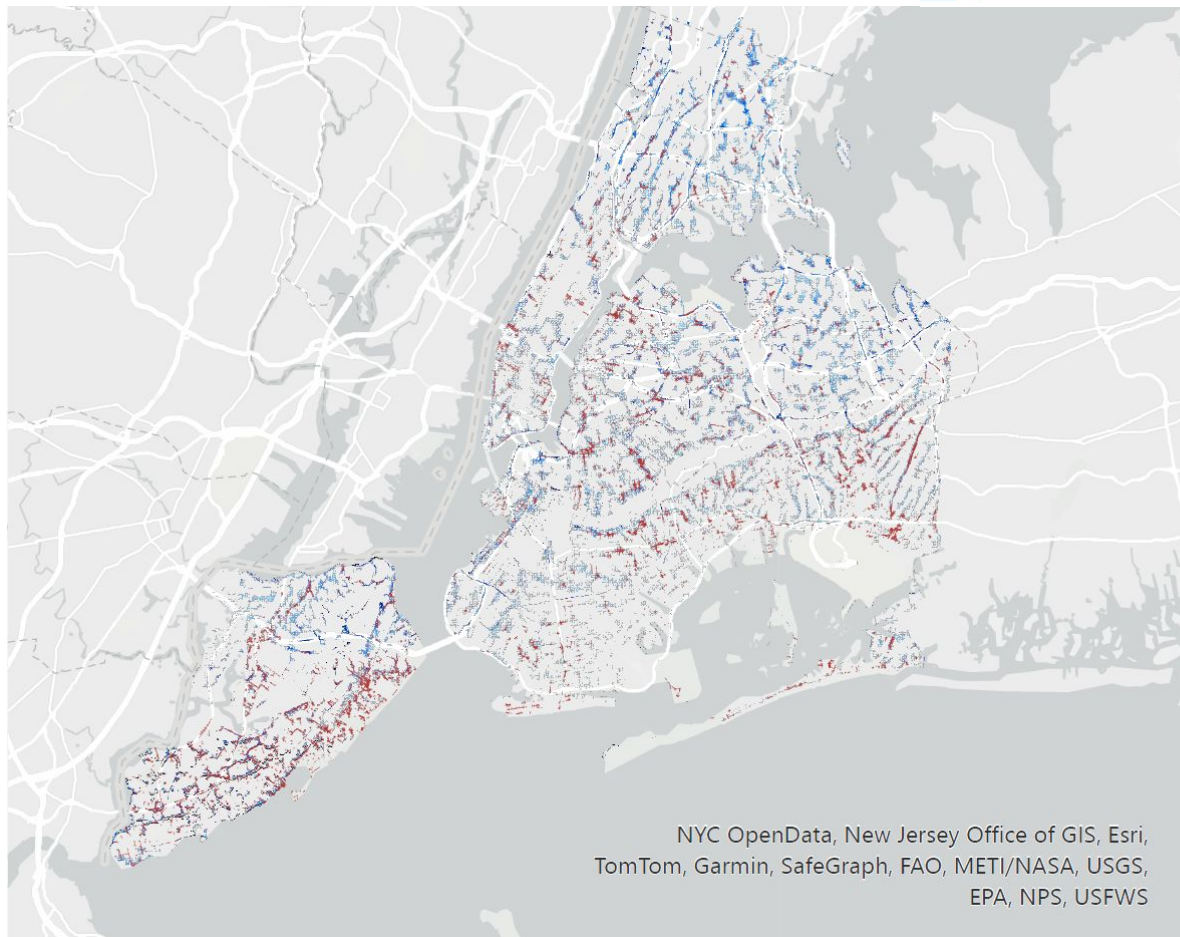
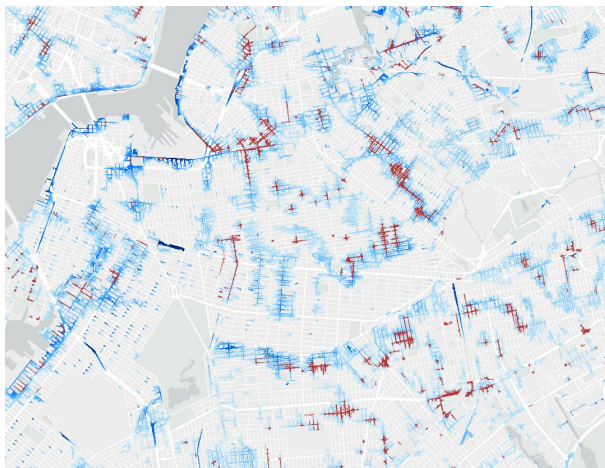
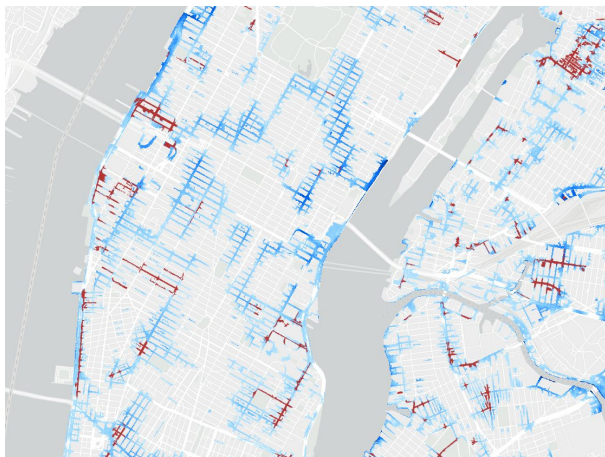
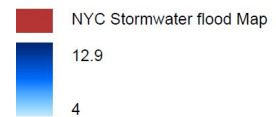
Number of USGS report point in study area	131
Number of point in predict flood zone	67(51.1%)
Total Run Time	1 h 40 min
Total hours of Ida event	26 hours
Resolution	10 feet (~ 3m)

Conclusion

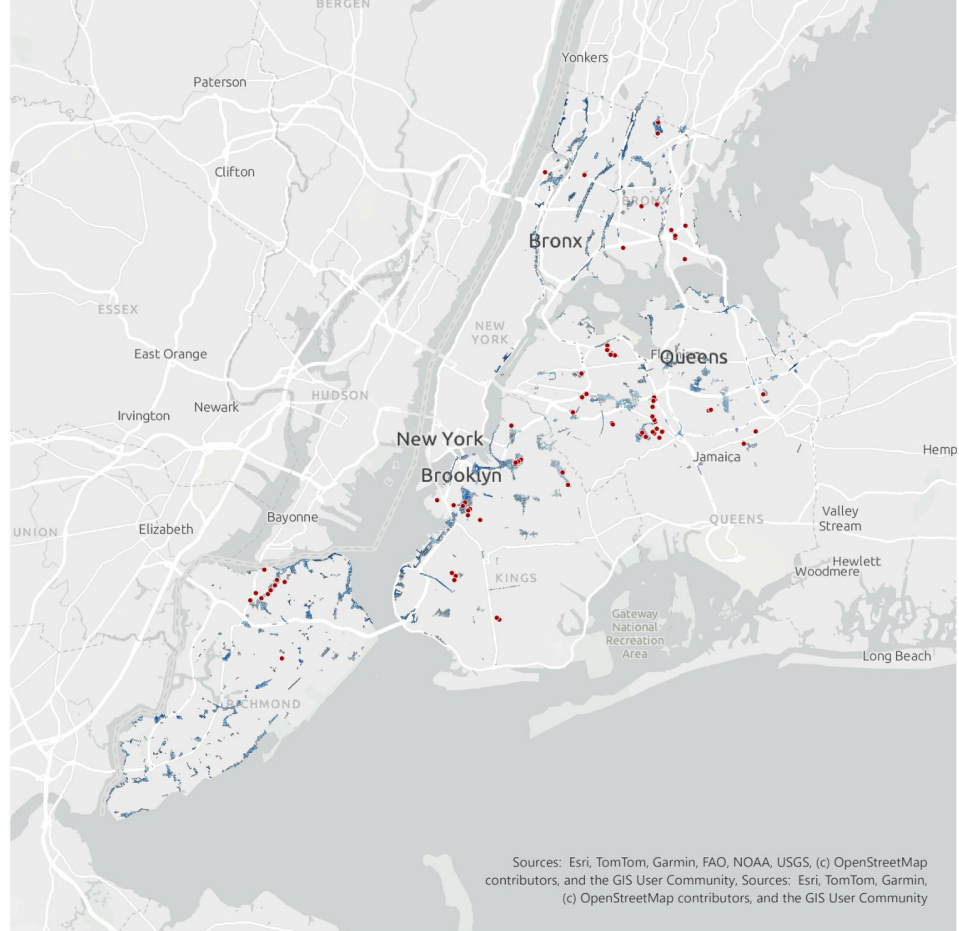
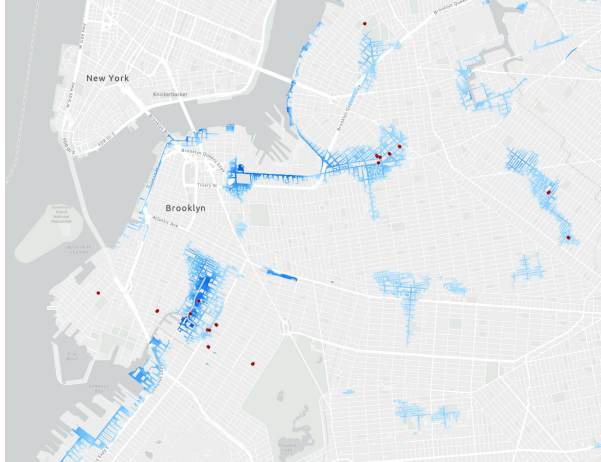
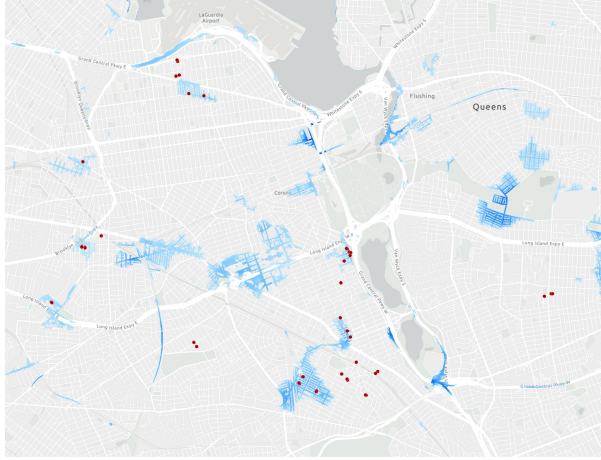
- NYC-scale, GPU-accelerated WCA2D model captures pluvial flooding at 3 m resolution in < 2 minutes, a 2× speed-up over CPU CA2D while retaining physical realism
- Integrated processes: non-uniform rainfall (kriging), SCS+GI runoff, dynamic Manning time–step, empirical sewer-drainage cap — together cut the false-alarm ratio by ~20 % with TPR ≈ 80 % under a 2.13-inch design storm
- Data-driven divisions algorithm collapses 106 k micro-basins into 135 hydraulically coherent units, keeping calculations tractable over 784 km²
- Hurricane Ida validation: rainfall RMSE = 2.26 in (48 h) across 29 gauges; spatial flood extents qualitatively match USGS report
- Actionable insight: platform rapidly tests Green-Infrastructure retrofits and drainage upgrades city-wide, guiding cost-effective risk-reduction.

Future Work

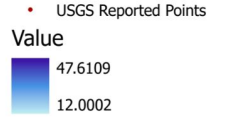




NYC OpenData, New Jersey Office of GIS, Esri,
TomTom, Garmin, SafeGraph, FAO, METI/NASA, USGS,
EPA, NPS, USFWS



Legend



Sources: Esri, TomTom, Garmin, FAO, NOAA, USGS, (c) OpenStreetMap contributors, and the GIS User Community, Sources: Esri, TomTom, Garmin, (c) OpenStreetMap contributors, and the GIS User Community



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Thank you !

Junyi Li